

Engineering Project

The Functions and Benefits of Shape Memory Alloy Actuators



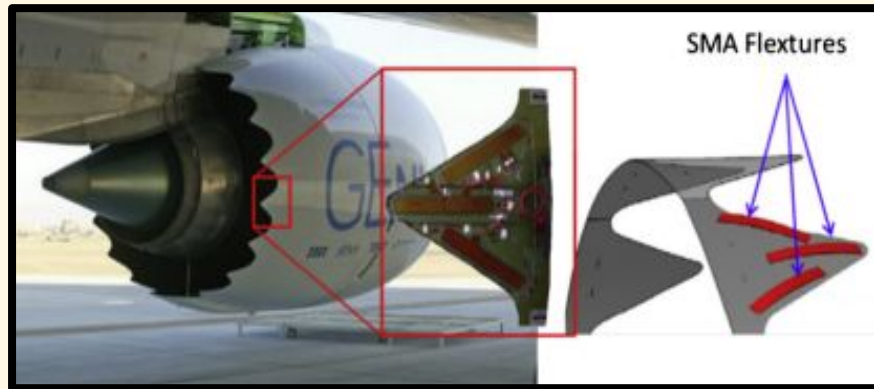
Picture Credits: <https://www.saesgetters.com/products-functions/products/shape-memory-alloys/shape-memory-alloys-nitinol-components>

Conducted By: Arian Salari

Quick Overview:

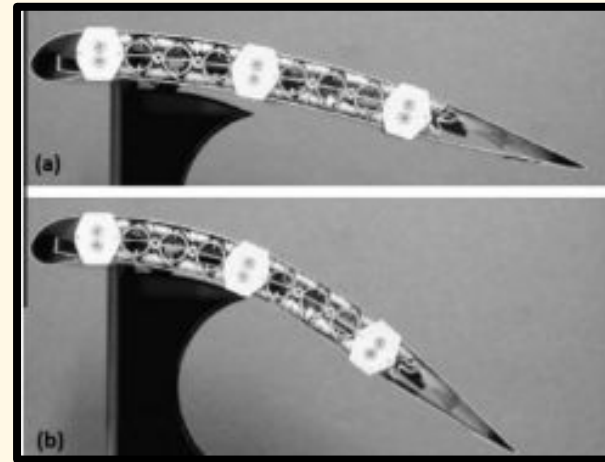
Shape-memory alloys are metals that even if they become deformed at below a given temperature, they will return to their original shape before deformation simply by being heated. Alloys with this unusual characteristic are used as functional materials in temperature sensors, actuators, clamping fixtures, and many other devices. Nitinol is one widely used shape-memory alloy consisting of nickel and titanium.

Boeing's variable geometry chevron:



<https://cpb-ap-se2.wpmucdn.com/uowblogs.com/dist/b/2950/files/2019/12/Review-1x5lz5z.pdf>

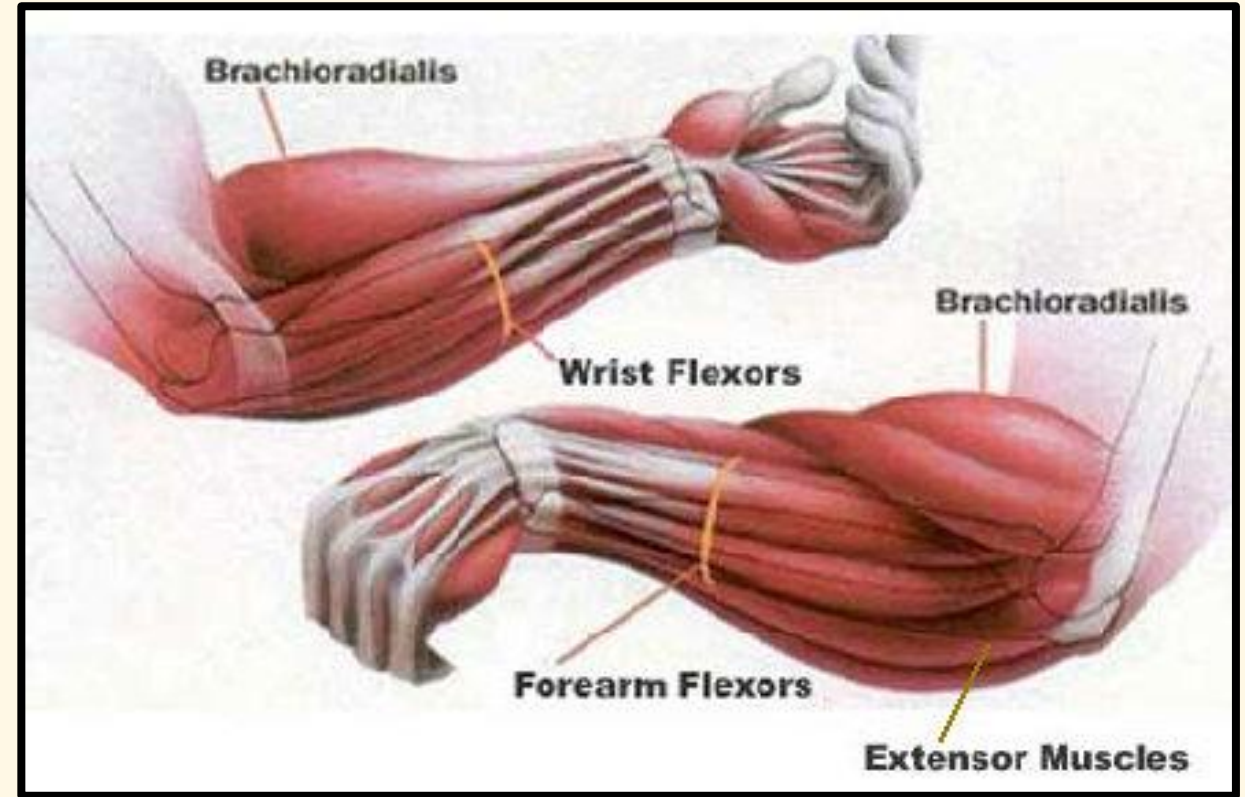
Wing morphing with antagonistic SMA:



<https://cpb-ap-se2.wpmucdn.com/uowblogs.com/dist/b/2950/files/2019/12/Review-1x5lz5z.pdf>

What's Being Tested:

Nitinol is known to have two main properties which are superelasticity and the shape memory effect. Throughout this project, the functionality of the shape memory properties of the nitinol alloy as an actuator are tested. The actuator is being used to repel and retract the fingers of the prototype prosthetic hand. It acts as a prosthetic counterpart to the muscles in the forearm known as the flexors muscles.



Picture Credits:

<https://therapyspot.ca/ot-tip-of-the-month-hand-strength-and-finger-dexterity/>

Materials:

Prosthetic Hand:

- Cardboard
- Straw
- String
- Gorilla Glue
- Tape
- Rubber Bands

Nitinol Spring Circuit (Prosthetic Muscles):

- Batteries
- Wiring
- Nitinol Springs
- Lever Switch

Not Required:

- Paint (used for aesthetic)
- Spray Paint (used for aesthetic)
- Wood or Cardboard Plank (used for stabilization of the arm)
- Glove (to cover hand for aesthetic)

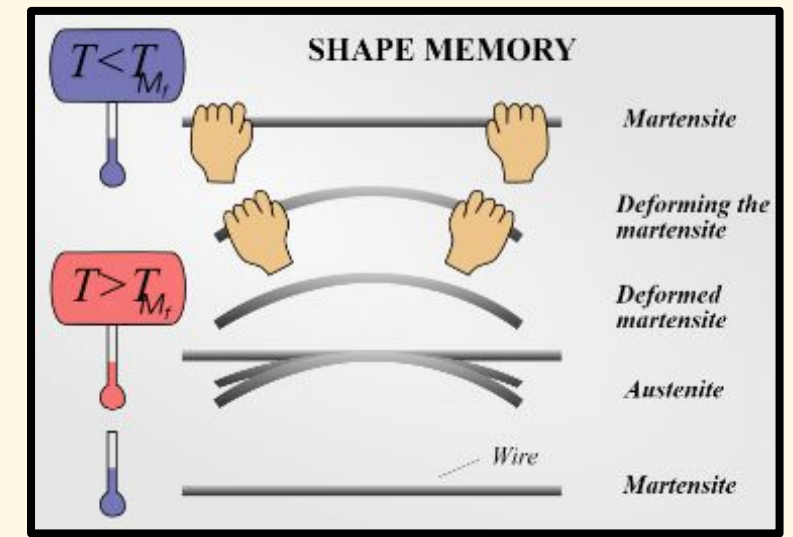


Picture Credits:

<https://www.worldconstructiontoday.com/news/construction-material-short-ages-to-continue-in-2021/>

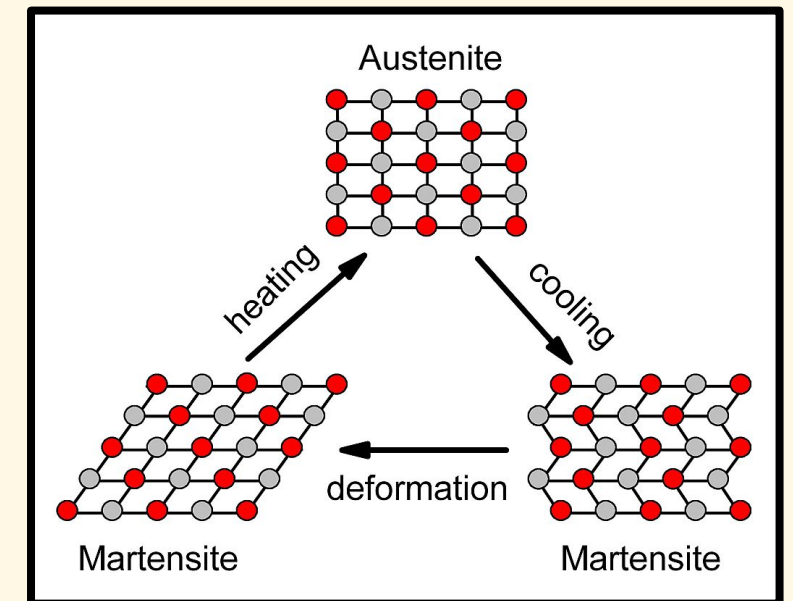
Discussion:

For a deeper understanding of this project, we need to review the process of shape shifting in shape memory alloys. First off, nitinol is a nickel titanium alloy (~50% Ni, ~50% Ti) which has two phases or crystalline structures, a high temperature (austenite) and a low temperature (martensite). As the low temperature phase is weaker, it is easier to bend and pull the material. When deformed at a low temperature and then heated, nitinol will return to the shape established when in the high temperature, stronger phase. By heating the material, the atoms are given enough energy to rearrange themselves back to their high temperature phase. The composition of the wire can be varied slightly to change the transformation temperature. This ability to remember and revert to the original shape gives this material the name “shape memory.” Some difficulties that were faced during this experiment were that for an actuator to work properly, the transition temperature from martensite to austenite needs to be above the present environments temperature. Setting this temperature is a trial and error process in which the spring is heated to a certain temperature over a certain time. Additionally, we first started with a single spring design to actuate all fingers but it couldn't overcome the artificial extensor (rubber band) force so we had to add individual springs for each finger.



Picture Credits:

<https://mstudent.com/what-are-shape-memory-alloys-metallurgy-how-they-work-and-applications/>

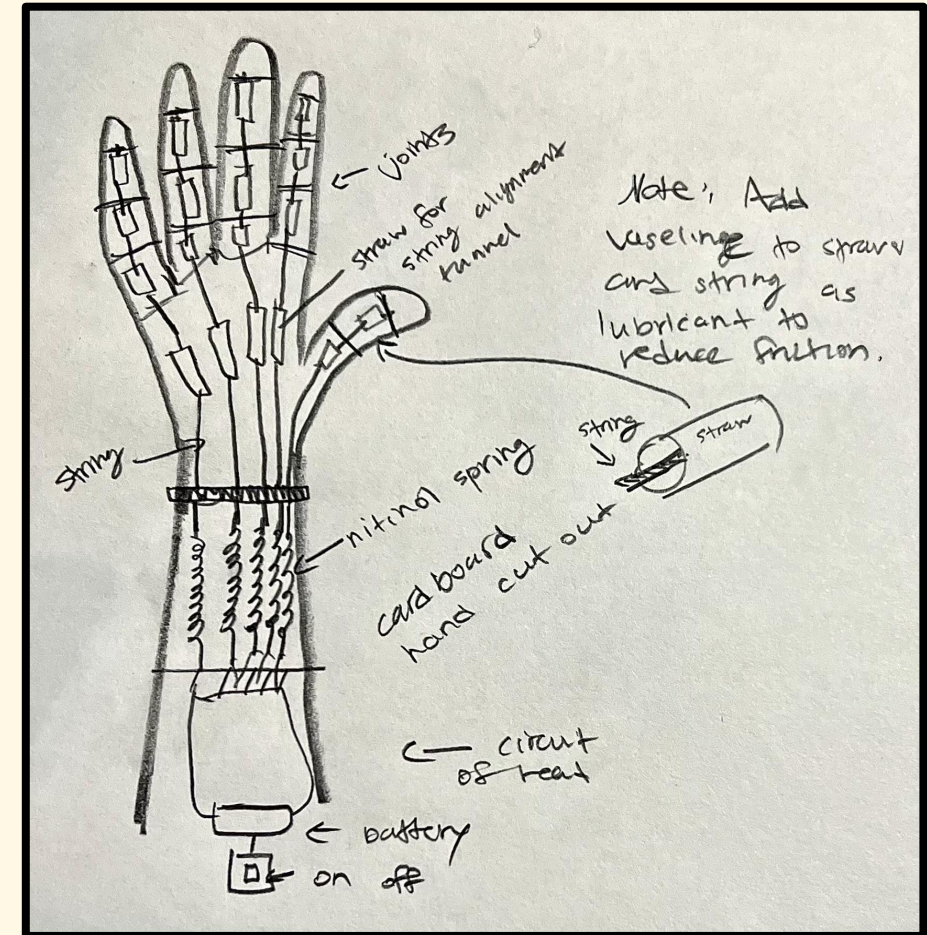


Picture Credits:

<https://ceramics.org/wp-content/uploads/2014/04/Shape-Memory-Alloys-Lesson.pdf>

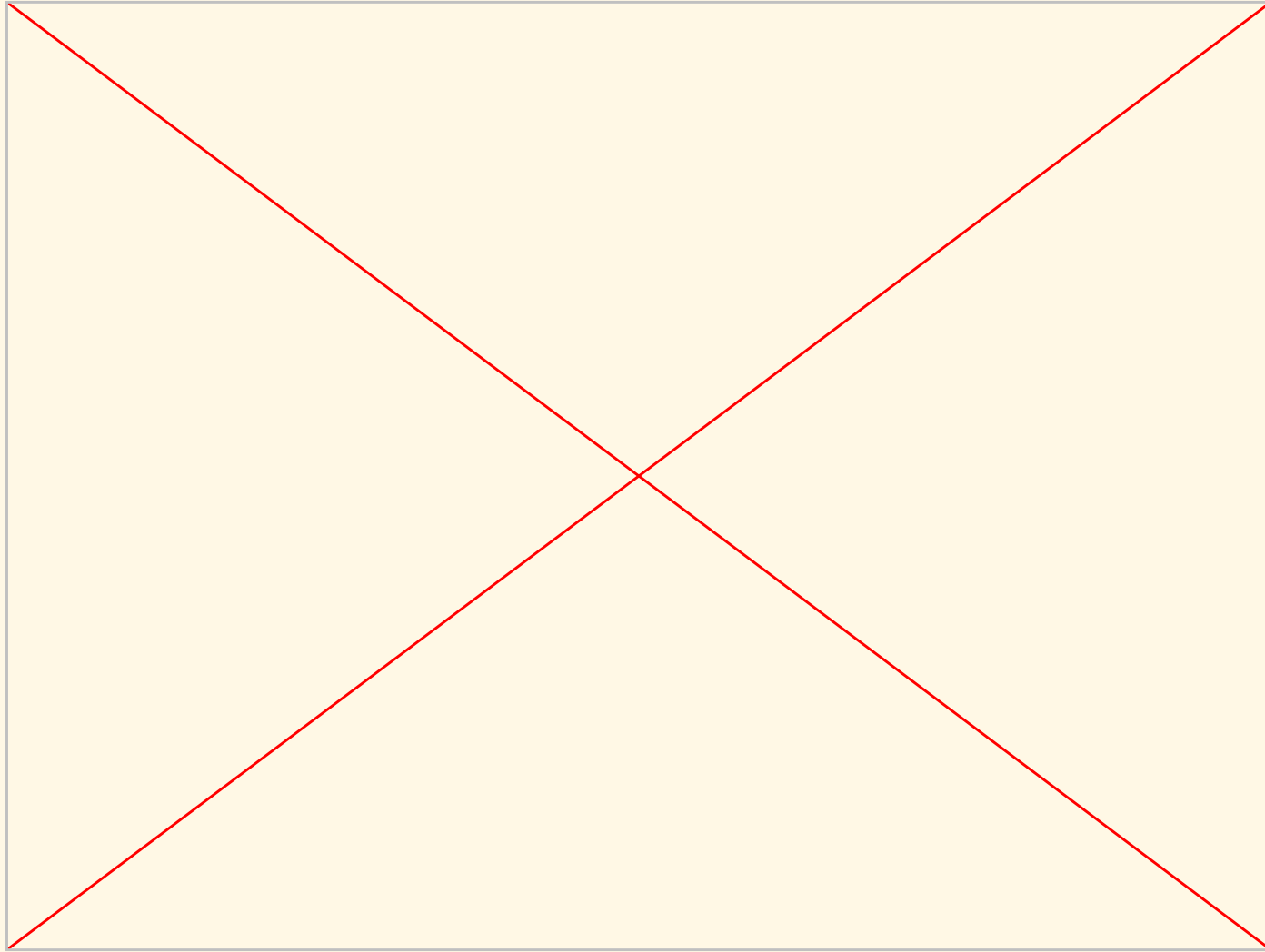
Prototype Model Design:

Before assembling the actual prosthetic arm, the idea and base of it was sketched out on paper in accordance with how a real human arm looks and functions. Before designing the arm, I studied the basic anatomy of a human arm to see which muscle groups, ligaments, and flexors help the fingers function. After studying the anatomy of a hand and arm and figuring out how it functions, I sketched out the prosthetic model on paper labeling the parts and how each part would function. After the sketch was finished I constructed the prosthetic arm out of the materials listed before. Our first set of springs were not generating enough force and therefore they were not able to fully actuate the fingers. Then, we shape set a stronger set of springs which were able to overcome the extensor force for each finger. After this, the shape memory actuator continued functioning perfectly as projected.



Prototype Model Sketch: This is the second prototype sketch and is the final revised design that the prosthetic arm was based off of.

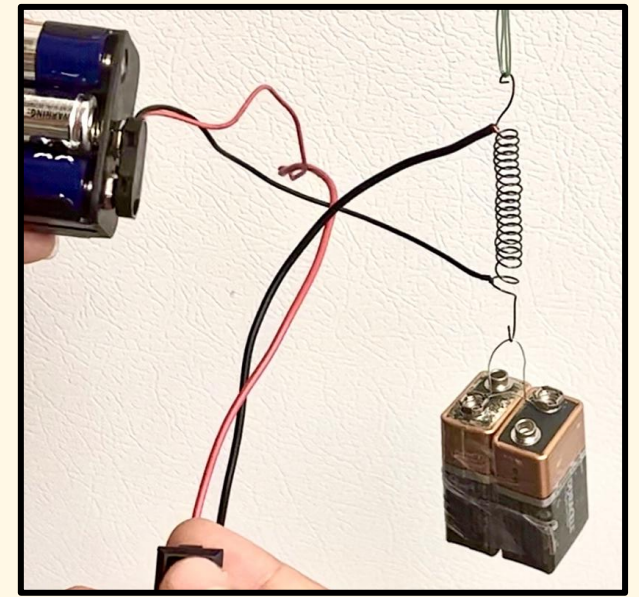
Prototype Functioning:



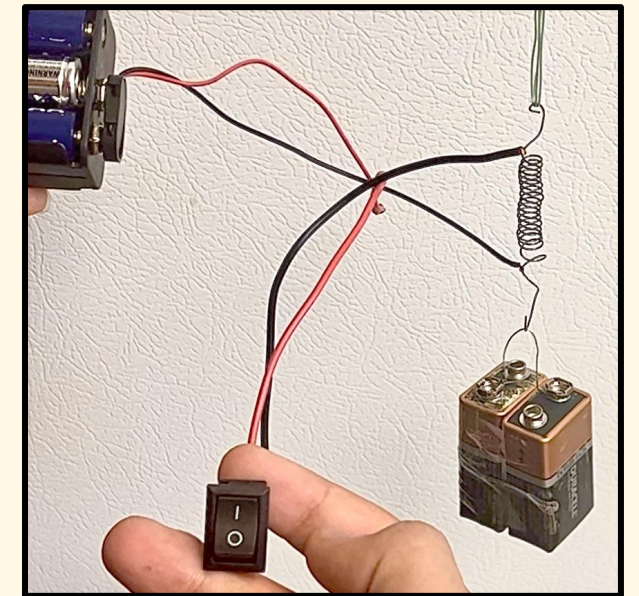
Final Prosthetic Hand: This is a video of the final prototype which portrays how the hand functions.

Results/Observations:

We first started with testing the nitinol spring's shape memory behavior. To this end I attached a weight to the spring to elongate it. Then I turned on the current. While the current was activated the temperature increased and the weight was lifted upwards indicating that the spring returned to its original shape. This is because it entered the austenitic phase. During the austenitic phase, nitinol achieves maximum stiffness. When turning the circuit off and cutting off the energy flow, the nitinol spring cooled down and returned to the elongated position. This is known as its martensitic phase which occurs when nitinol goes back to lower temperatures below the transition temperature. Having used this same concept within the prosthetic hand, the nitinol's shape-memory properties are able to make the hand function. When the spring tightens up during its austenite phase the fingers curl into a fist representing flexed muscles and when the fingers ease and go back to the relaxed position it means that the spring is in the martensitic phase representing a relaxed hand with no muscle action.



Martensite Phase: No energy flow and nitinol spring is at ease (No muscle action/fingers at ease)



Austenite Phase: Energy is flowing and the spring tenses up (Muscles flexed/ fingers curled)

Reflection/Application:

Throughout conducting the project, I learned many new things that I wasn't aware of before such as the fact that medical grade nitinol is mostly used for its superelasticity component. We used medical grade nitinol for its shape-memory property in this case and it still functioned fine. Something that I may have done differently was the design of the flexor muscles of the prosthetic arm. Instead of using 5 individual different nitinol strings for the arm's muscle I could have tried to make one strong singular powerful spring connected to all fingers. The next step for researching this problem would most definitely be finding the perfect temperature range to make the spring function the most effectively. Finding this out would allow us to provide the right amount of heat needed every single time for a specific amount of retraction needed in the fingers. The results of this project can be applied to our everyday lives as the results of this prosthetic arm draw conclusions such as the fact that with simple technology we can make highly advanced performing objects like that of the human hand which can perform plentiful tasks. These results can also be applied to other studies as the findings of nitinol's functions and the way it works can be used in other projects to make other tasks more efficient as well as nitinol's many properties can prove useful for countless tasks.

Picture Credits:

<https://space.stackexchange.com/questions/23758/are-these-experimental-s-toichiometric-niti-wire-mesh-wheels-made-from-memory-al>



Nitinol Rover Wheels: Have proven very capable and beneficial for missions in space.

Picture Credits:

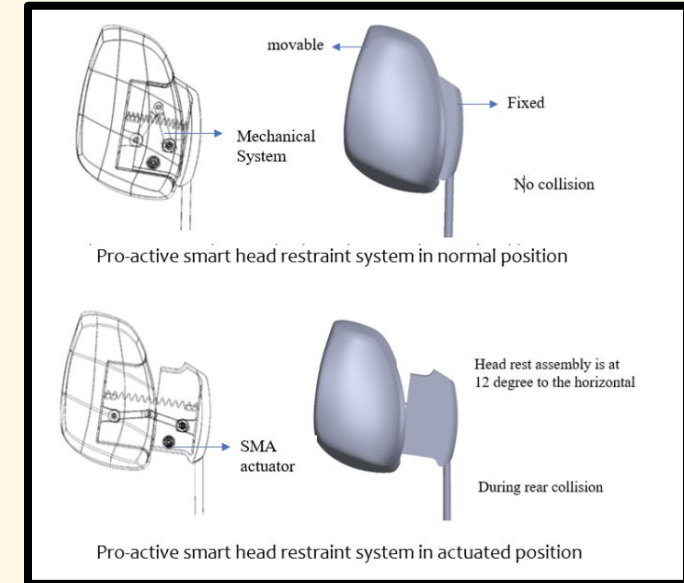
<https://technology.nasa.gov/patent/LEW-TOPS-124>



Nitinol Winglets: Nitinol's superelasticity and shape memory alloys winglets to function more efficiently.

Conclusion:

The shape memory properties of nitinol were used to actuate a prosthetic hand. Nitinol springs were tuned such that their transition temperature was above the room temperature. Therefore, at the room temperature the springs were at the martensitic phase. Throughout the experiment, it was proven that when the circuit was activated and the spring started to heat up above the transition temperature, the nitinol spring entered the austenitic phase and went back to its shape set configuration. During the austenitic phase, nitinol achieves maximum stiffness. When turning the circuit off and cutting off the energy flow, the nitinol springs went back to the martensitic phase. Considering that these functions were implemented into a prosthetic hand that was able to successfully perform and function like a basic human hand, this project proves that nitinol is beneficial and important as it can carry out unique functions that others metals can't while also being fairly efficient to work with and simple to make. Overall, it can be concluded that shape memory alloy actuators are beneficial and efficient as they can perform tasks that may usually need way more technology to perform but due to its natural properties nitinol can partake in the completion of these tasks.



Picture Credits:

<https://contest.techbriefs.com/2020/entries/automotive-transportation/10721>

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- Simone, F., Rizzello, G., Seelecke, S., & Motzki, P. (2020, November 17). *A soft five-fingered hand actuated by shape memory alloy wires: Design, manufacturing, and Evaluation*. Frontiers. Retrieved December 26, 2022, from <https://www.frontiersin.org/articles/10.3389/frobt.2020.608841/full>
- *Shape memory alloys - american ceramic society*. (n.d.). Retrieved December 26, 2022, from <https://ceramics.org/wp-content/uploads/2014/04/Shape-Memory-Alloys-Lesson.pdf>
- *Applications of shape memory alloys in the medical field*. Applications of Shape Memory Alloys in the medical field :: Total Materia Article. (n.d.). Retrieved December 30, 2022, from <https://www.totalmateria.com/page.aspx?ID=CheckArticle&site=ktn&NM=212>

Videos/ Documentaries:

- <https://www.youtube.com/watch?v=Pn-6bGORy0U>
- https://www.youtube.com/watch?v=bvw7_a2gU24
- <https://www.youtube.com/watch?v=wI-qAxKJoSU>

Log #1

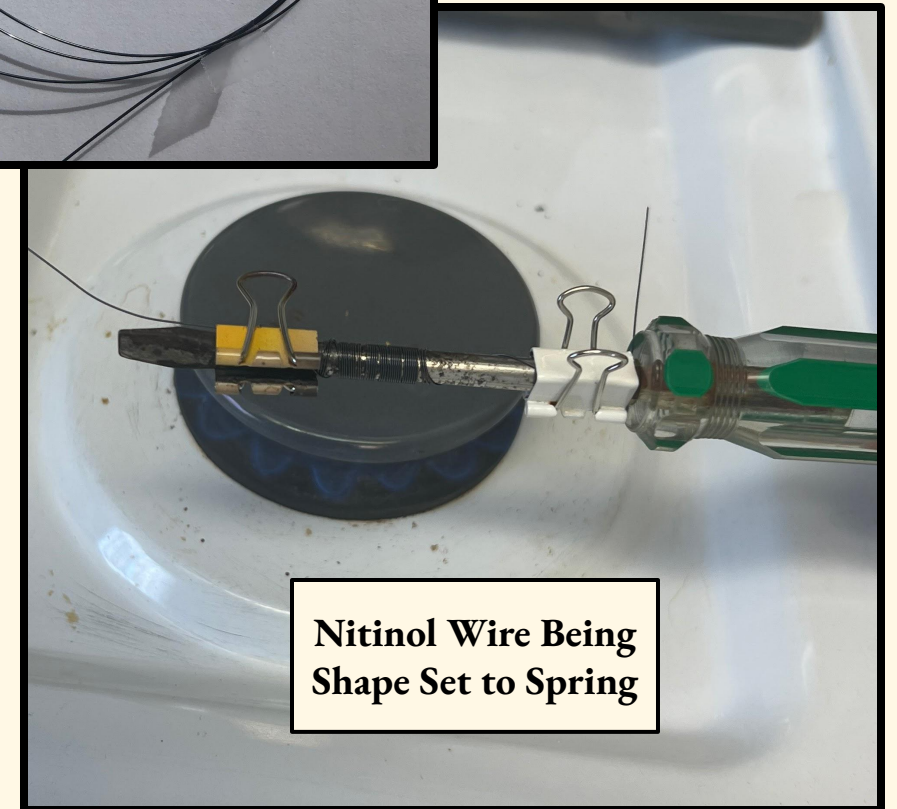
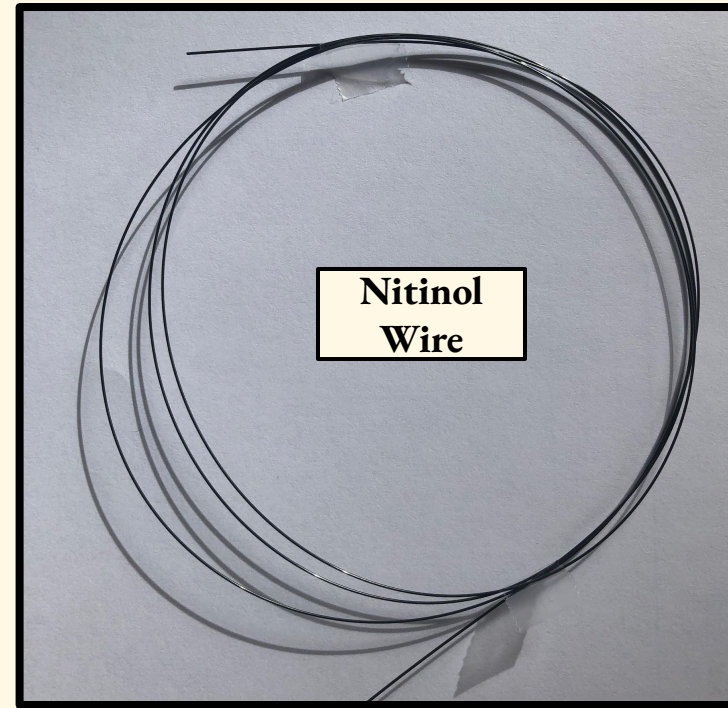
Having competed in the 2021 Orange County Science and Engineering Fair during freshman year, I also wanted to compete again this year to see how far I could make it this time. After visiting the Edwards Lifesciences campus, I became quite interested with biomechanics and came up with the idea of this project. After presenting the idea to my instructor Ms. Mattair who was fascinated by it, she pointed out the guidelines and a few things that I would need to consider for this project.

Fluids / Aerodynamics		
JUNIOR DIVISION	<i>First Place</i>	Hussein Hirani
	<i>Second Place</i>	Hannah Saab, Lucas Saab, Jocelyn Inavolu Josiah Harris
	<i>Third Place</i>	Arjun Appa, Dev Batra and Ryan Colley (team)
SENIOR DIVISION	<i>First Place</i>	Firas Qureshi
	<i>Second Place</i>	Arian Salari Alex Liu
	<i>Honorable Mention</i>	Daichi Hiraoka Madeleine Yee

Picture Credits: <https://ocsef.org/2021/>

Log #2

After being approved to pursue with this project, I immediately got to work and began gathering the main materials and putting them to the test. First I started to design the arm on paper then mapped out the main components. After all this I tested the nitinol's shape-memory property by shape setting it to the shape of a spring. It was very interesting to me and broadened my understanding of how the shape setting process actually works.



Log #3

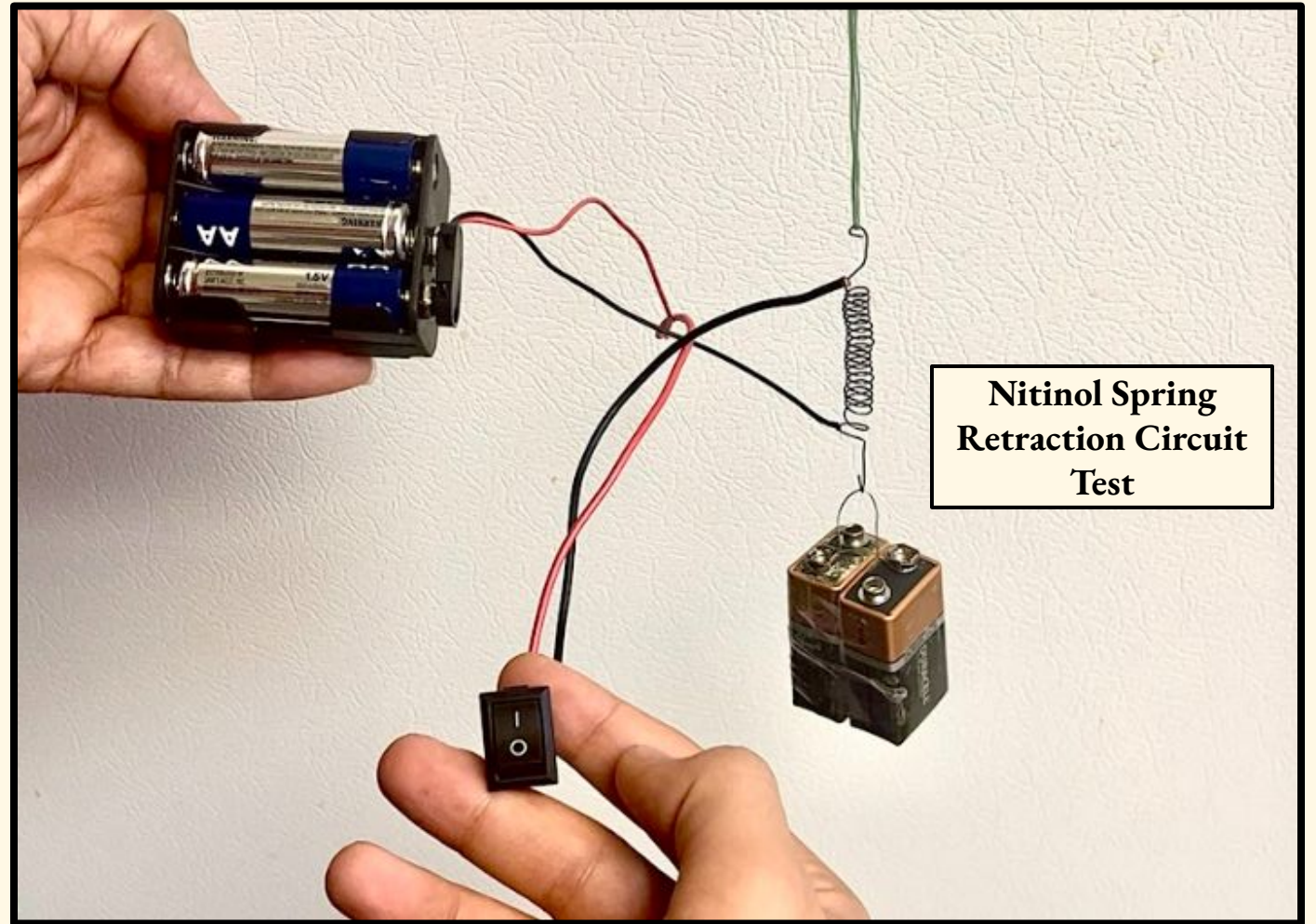
After creating a test spring out of a 8 inch long strand of nitinol wire, we put together the hand which would function based off of the shape-memory property of the nitinol. The hand is a real sized cardboard cut out hand bent at the joints for maximized flexibility just like a real hand is structured. The cut straw tubes are used to stabilize the strings which act as prosthetic tendons stabilizing the fingers and would be attached to the nitinol springs acting as the flexor muscles.



Cardboard Cut
Out Prosthetic
Hand

Log #4

Before implanting the circuit into the prosthetic cardboard hand and creating the final product, we tested it to make sure that it works accordingly. The circuit worked just as planned as when the switch was turned on and the spring was heated up, it tightened up meaning that the shape-memory property of the nitinol had reached the austenite stage representing a tightened flexor muscle which would act the same way.



Log #5

After all testing and procedures turned out successful, we implanted the nitinol spring circuit systems into the cardboard arms forearm which substitutes for the functions of a real human hands flexor muscles located in the forearms. This is the muscle group that mainly controls the hand and fingers movements. The prosthetic hand turned out to be successful further proving that new and simplified technology such as that of nitinol's functions can prove beneficial and successful.

